

A Collection-based Digital Product Passport Suitable for Tracking Parts With No ID

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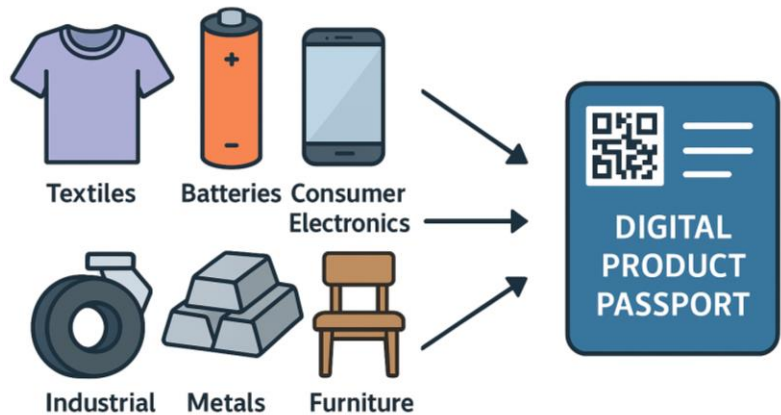
smartFactory KL®

dfki Innovative
Fabriksysteme
ai

ISM 2025
International Conference on
Industry of the Future and Smart Manufacturing

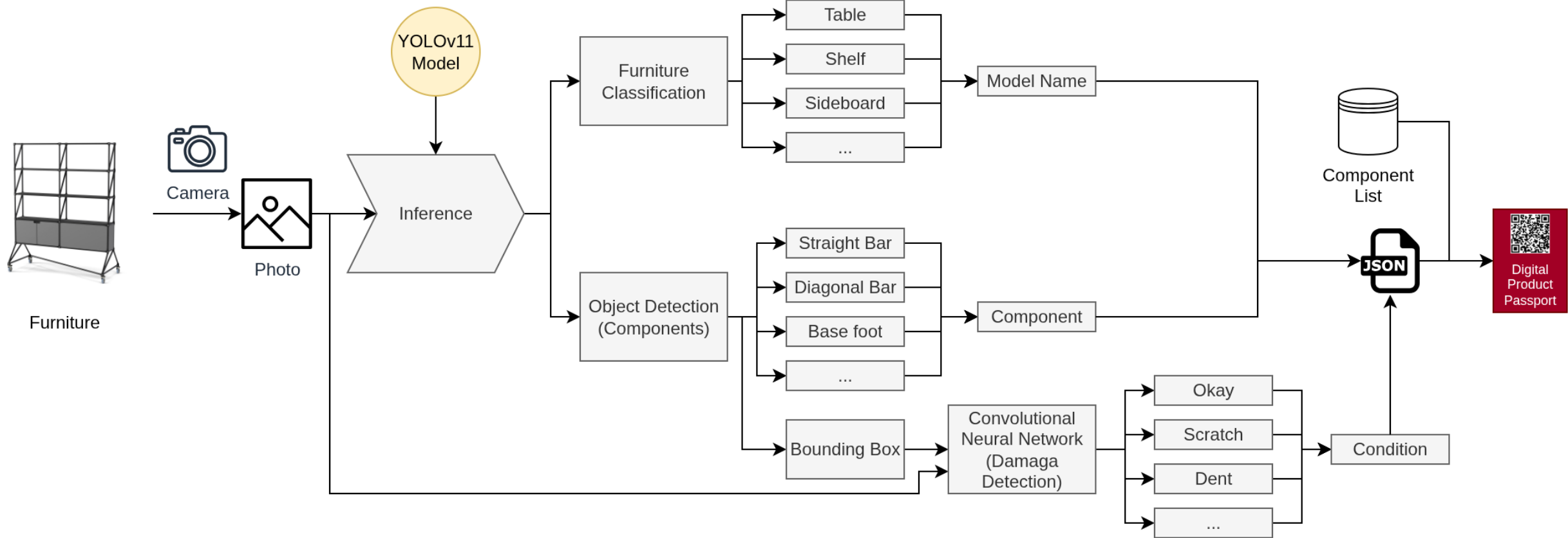
Motivation

- Ecodesign for Sustainable Products Regulation (ESPR) will require Digital Product Passports (DPPs) for many product groups like textiles, batteries, consumer electronics and furniture
- DPPs will contain information on sustainability performance, repairability, and end-of-life information
- European Commission's approach to more environmentally sustainable and circular products.
- System180 builds modular furniture through stainless tube systems designed for 20+ years of reuse



Solution Concept: Collections

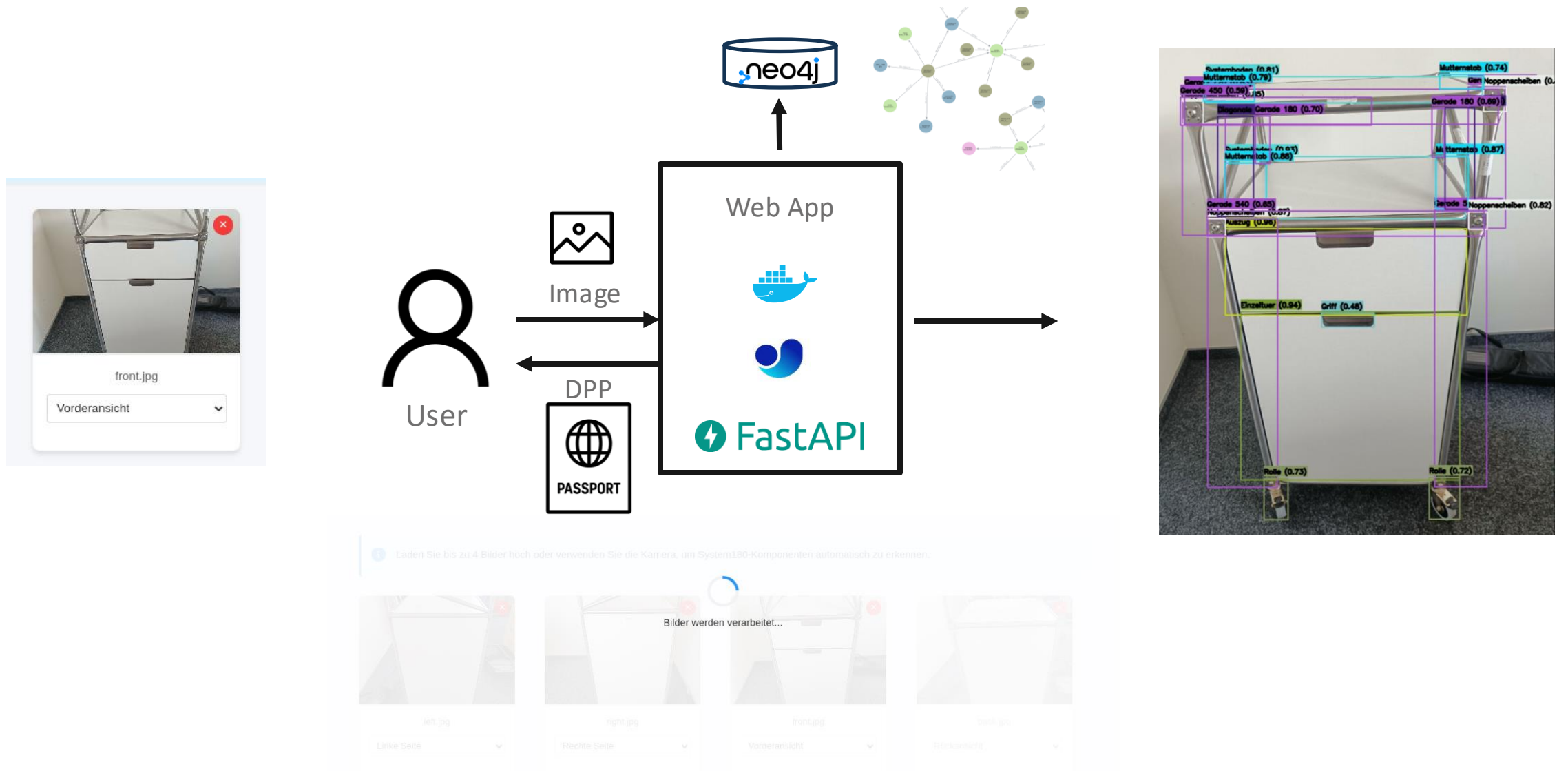
- YOLOv11 models trained on synthetic data (NVIDIA Isaac Sim) and labeled photos



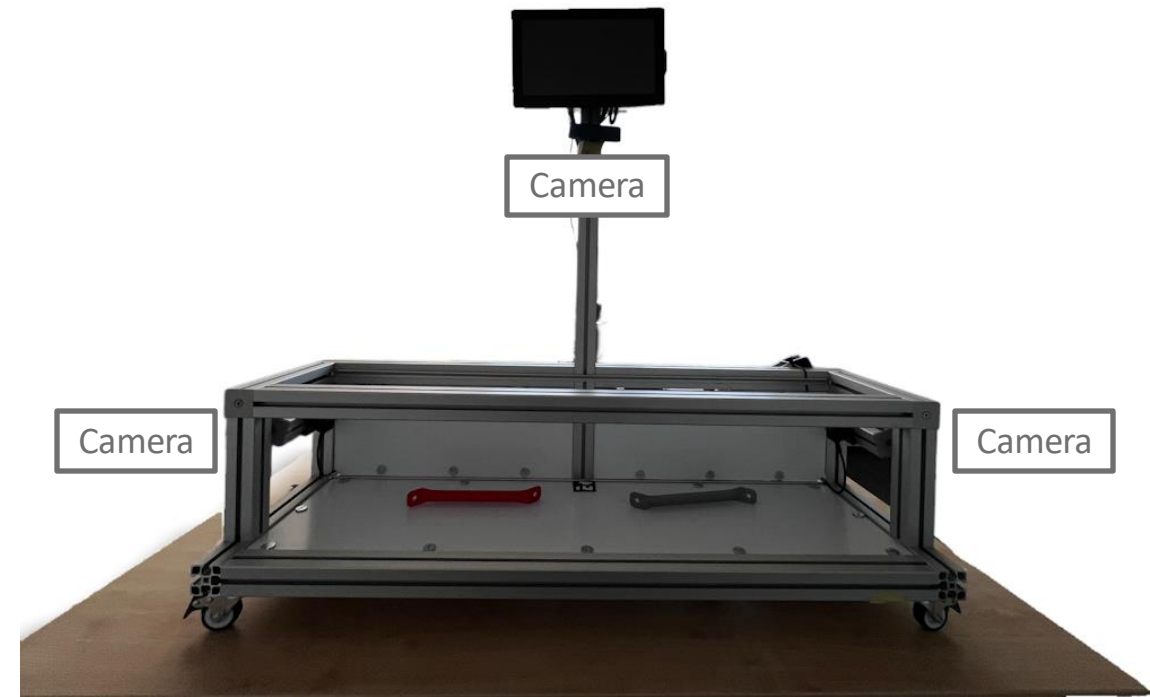
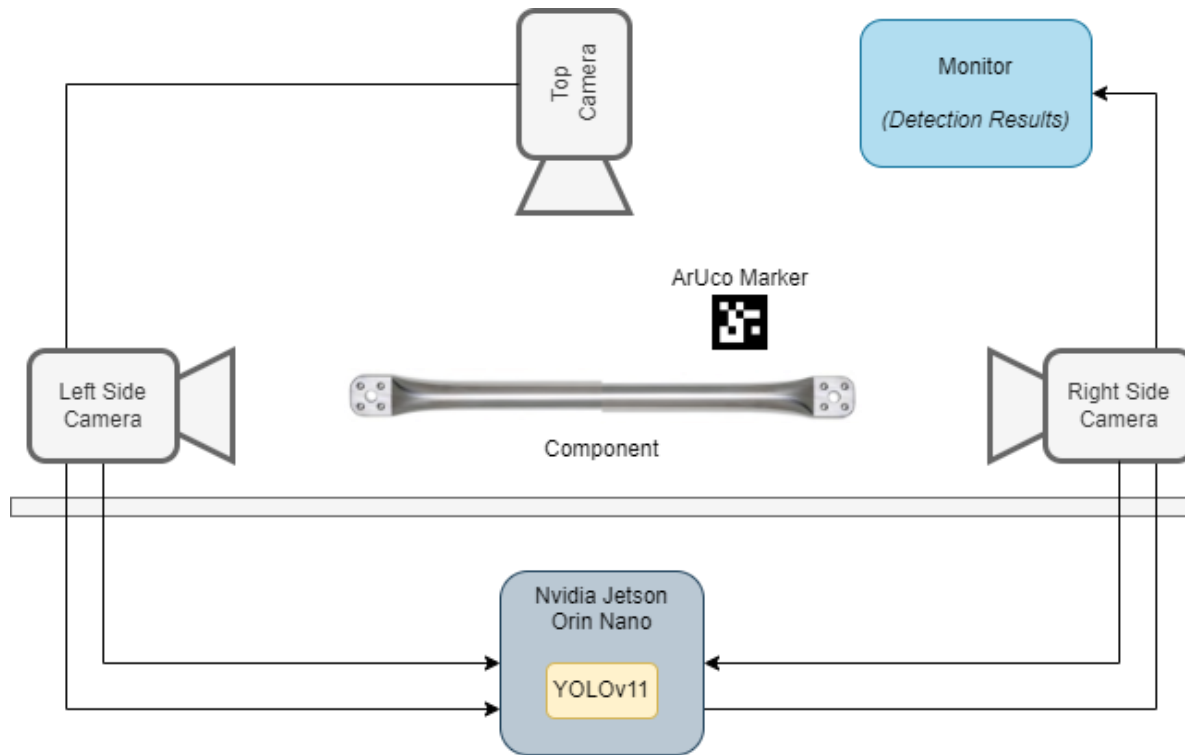
Training

- YOLOv11 (real photos)
- YOLOv11 (synthetic data)
- fused via IoU voting
- Custom CNN for condition (okay, scratch, dent).
- Color & coating via K-Means palettes
- ≈10k Isaac-Sim renders + ≈1k labeled factory photos (Label Studio workflow).

Technical Implementation: Web Application



Demonstrator



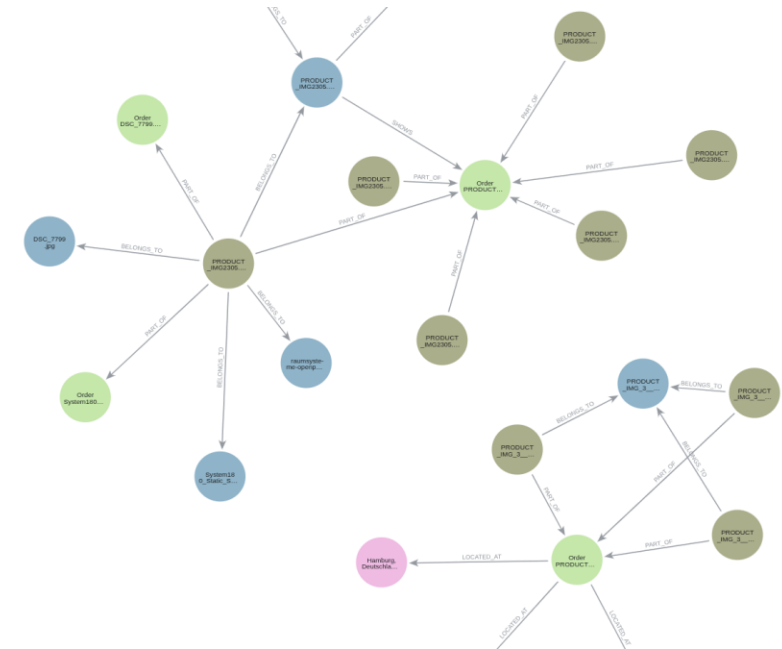
Results

Detection & Measurement

- 9 component classes reliably detected in browser uploads and edge rig.
- Measurement heuristics map pixels to the nearest catalogue length (180–900 mm) using known references.
- Condition CNN surfaces scratches/dents before operators decide on reuse.

Systemboden	87%
Gerade 360	87%
	
ANSICHT right	AUFNAHMETYP symmetric
MASSE 20 x 360 mm	TYP Strebe
GEWICHT 150-300 g	FARBE Edelstahl 64% HEX: #C0C0C0 NCS: 2500-N

Noppenscheiben	89%
Mutternstab	89%
Gerade 360	88%
Noppenscheiben	87%
Gerade 720	84%
	
ANSICHT back	AUFNAHMETYP symmetric
MASSE 20 x 720 mm	TYP Strebe
GEWICHT 150-300 g	FARBE Edelstahl 90% HEX: #C0C0C0 NCS: 2500-N
ZUSTAND unbeschädigt Wiederverwendbar	
Unbeschädigt KI-Detektion Melden	



Limitations

- **Condition Detection:**
 - Scratches and small dents are hard to see in RGB photos.
 - Condition CNN still need human verification
- **Multi-view fusion:**
 - Each view is stored separately, no automatic algorithm merges front/back/left/right detections into a single 3D-model yet
- **Edge performance:**
 - Jetson Orin throttles when four models run simultaneously;
 - Converted models to TensorRT .engine files and reduced them to FP16 to stay within power/thermal budgets.

Conclusion

- Camera-based, ID-free workflow delivers the core DPP data for modular furniture (class, length band, décor, condition).
- Collection Principle in Neo4j ensures every component belongs to exactly one order/collection, giving a continuous trace over reuse cycles.
- YOLOv11 ensemble + measurement heuristics + colour and condition pipelines populate these attributes consistently in edge demonstrator.
- Populated Neo4j graph, dashboards and CSV exports provide concrete evidence that the approach runs on real System180 data today.

Outlook

- Improve defect capture with better optics/datasets so the condition model can reliably see very small scratches and dents.
- Develop multi-view fusion to combine front/back/left/right detections into a single 3D-consistent component representation in the graph.
- Introduce multi-marker photogrammetry on the rig and more robust length estimation in the web workflow.
- Map the Neo4j schema to Asset Administration Shell (AAS) / Ontology Design Pattern vocabularies and add an ESPR-compliant DPP export service.
- Add furniture-level classification and incremental/on-device learning so new components and variants can be integrated without full retraining.

Thank you for your Attention

- Contact: peter.stein@dfki.de
- GitHub: <http://github.com/Green-AI-Hub-Mittelstand/SUSPEKT>

Time for Questions

Sources

- https://commission.europa.eu/energy-climate-change-environment/standards-tools-and-labels/products-labelling-rules-and-requirements/ecodesign-sustainable-products-regulation_en



Screenshots

Intelligente Objekterkennung

für System180-Komponenten

i Laden Sie bis zu 4 Bilder hoch oder verwenden Sie die Kamera, um System180-Komponenten automatisch zu erkennen.



left.jpg

Linke Seite



right.jpg

Rechte Seite



front.jpg

Vorderansicht



back.jpg

Rückansicht



Wählen Sie den Aufnahmetyp aus:

☒ **Symmetrisch** (benötigt nur Vorder- und eine Seitenansicht)

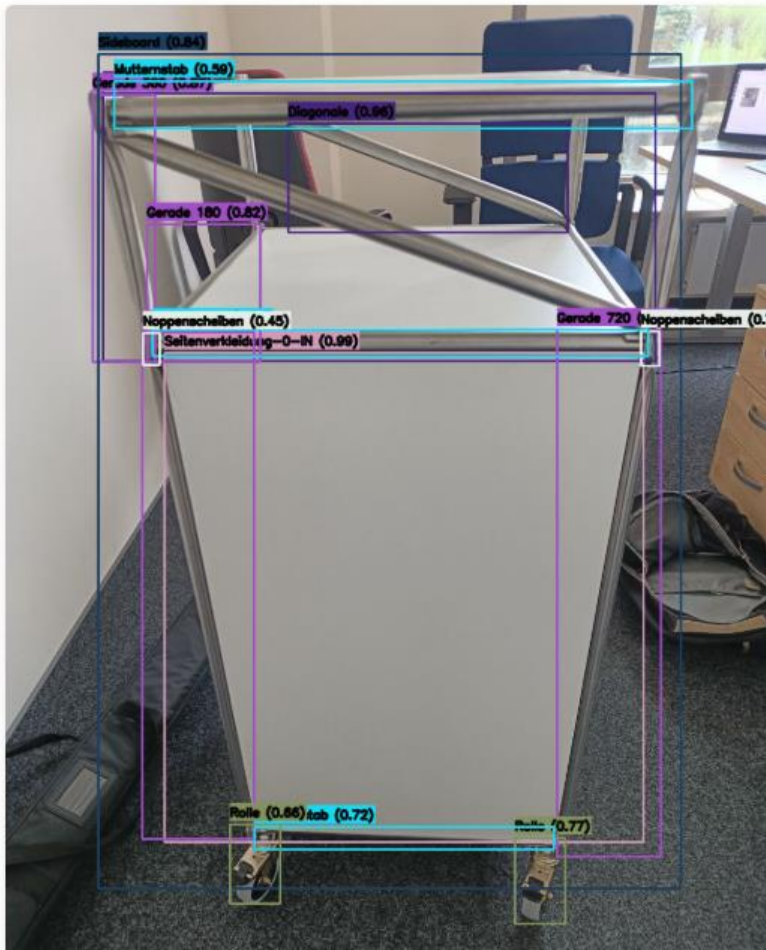
☐ **Asymmetrisch** (benötigt Ansichten von allen Seiten)

☐ **Einzelaufnahme** (Ansicht ist egal)

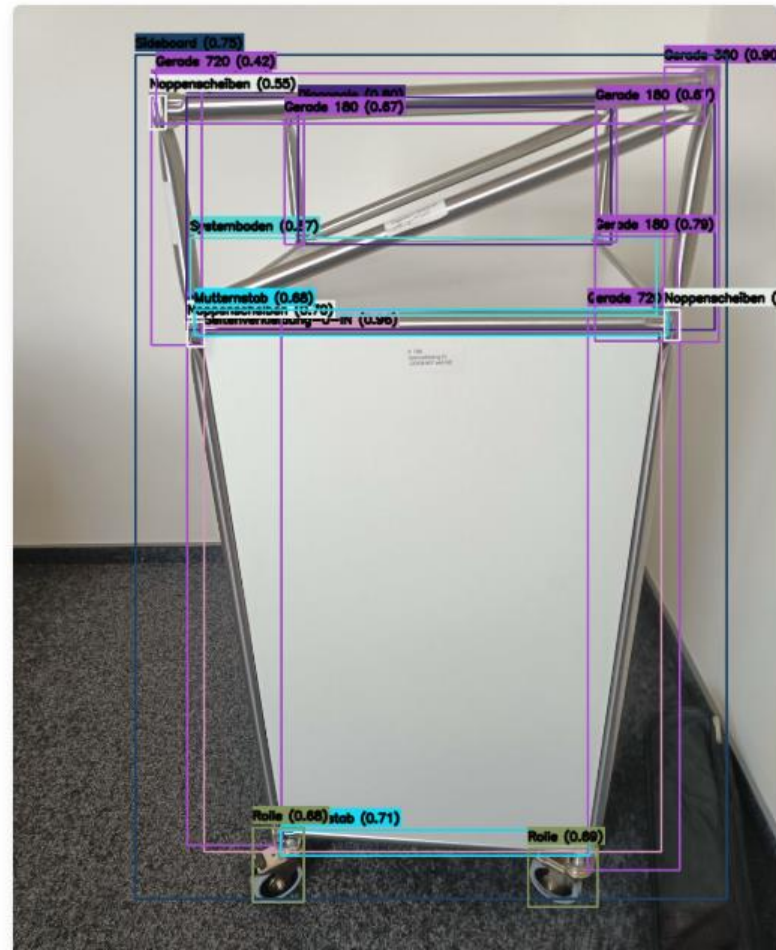
Erkennungsergebnisse

System180-Komponenten erfolgreich analysiert

left.jpg



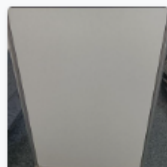
right.jpg



Erkannte Objekte

Seitenverkleidung-0-IN

99%



ANSICHT
left

AUFNAHMETYP
symmetric

MASSE
None

TYP
Verkleidung

GEWICHT
450-650 g

FARBE

 Flieder 90%

HEX: #b699bd

NCS: 2030-R50B

ZUSTAND

unbeschädigt Wiederverwendbar

Unbeschädigt ▼

KI-Detektion

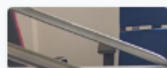
 Melden

Diagonale

98%

Diagonale

96%



ANSICHT
left

AUFNAHMETYP
symmetric

MASSE
None

TYP
Strebe

Erkannte Objekte

Seitenverkleidung-0-IN

96%



ANSICHT
right

AUFNAHMETYP
symmetric

MASSE
None

TYP
Verkleidung

GEWICHT
450-650 g

FARBE

 Lichtgrau 99%

HEX: #c8c8c7

NCS: 2000-N

ZUSTAND

unbeschädigt Wiederverwendbar

Unbeschädigt ▼

KI-Detektion

 Melden

Diagonale

91%

Gerade 360

90%

Gerade 720

89%

Gerade 720

87%

Systemboden

87%



Inventarübersicht

Ressourceneffizienz & Bestandsmanagement

Bestandsdaten

Aktualisiert: Heute

Inventarübersicht

Auftrag	Komponente	Material
Filter	Filter	Filter
PRODUCT_IMG2305.jpg_20250324	Doppeltuerblatt	{"confidence": 0.97, "erkannte_farbe": "Wei\u00df", "he
PRODUCT_IMG2305.jpg_20250324	Doppeltuerblatt	{"confidence": 0.97, "erkannte_farbe": "Wei\u00df", "he
PRODUCT_IMG2305.jpg_20250324	Doppeltuerblatt	{"confidence": 0.97, "erkannte_farbe": "Wei\u00df", "he
PRODUCT_IMG2305.jpg_20250324	Doppeltuer	{"confidence": 0.97, "erkannte_farbe": "Wei\u00df", "he
PRODUCT_IMG2305.jpg_20250324	Doppeltuer	{"confidence": 0.97, "erkannte_farbe": "Wei\u00df", "he
PRODUCT_IMG2305.jpg_20250324	Doppeltuer	{"confidence": 0.97, "erkannte_farbe": "Wei\u00df", "he
PRODUCT_IMG2305.jpg_20250324	Doppeltuerblatt	{"confidence": 0.97, "erkannte_farbe": "Wei\u00df", "he
PRODUCT_IMG2305.jpg_20250324	Griff	{"confidence": 0.89, "erkannte_farbe": "Edelstahl", "hex

SYSTEM 180 – NACHHALTIGKEIT IM FOKUS

Ressourceneffizienz – Analyse wiederverwendbarer Komponenten



Gesamtlänge

4283.1 m



Wiederverwendbare Länge

2979.1 m



Gesparte Masse

734.69 kg



Materialeinsparung

69.55 %



CO₂ Einsparung Produktion

-912.79948 kg



TRANSPORTEMISSIONEN IM ÜBERBLICK



Gesamtstrecke

15865.66 km



Online-Bestellungen

12289.97 km



CO₂-Emissionen Gesamt

12851.19 kg



CO₂-Emissionen Online

2120.02 kg

● Online-Bestellungen ● Vor-Ort-Bestellungen ● Transportrouten



